H2 Math Summary

Example Questions

Maximus Pung

Contents

1 Systems of Linear Equations 2

Question 1.1 Forming equations from a table ... 2
Question 1.2 Unknown coefficients of a polynomial ... 3
Question 1.3 Unknown constants in a recurrence relation ... 4
Question 1.4 More equations than variables ... 5
Question 1.5 Fewer equations than variables with an inequality ... 6

2 Inequalities 7

Question 2.1 Graphical method and sign test ... 7
Question 2.2 Sign test and $\frac{1}{x}$ substitution ... 8
Question 2.3 Modulus inequality and substitution ... 9
Question 2.4 Modulus graphs and area ... 10

3 Graphing Techniques 11

Question 3.1 Range of values y cannot take and rational curve sketch ... 11
Question 3.2 Stationary point condition and line through intersection of asymptotes ... 13
Question 3.3 Increasing rational function and superimposed hyperbola ... 15
Question 3.4 Parametric sketch with endpoint and intersection with another curve ... 17
Question 3.5 Cartesian equation of parametric curve and its oblique asymptotes ... 19
Common Question Types ... 21

4 Graphing Transformations 22

Question 4.1 Reverse transformation to find the original equation ... 22
Question 4.2 Translation, scaling, modulus and reflection sketches ... 24
Question 4.3 Graphs of $\mathrm{f}'(x)$, $\frac{1}{\mathrm{f}(x)}$ and $|\mathrm{f}(x)|$ from $y = \mathrm{f}(x)$ ... 26
Question 4.4 Rational graph and $y = \frac{1}{x}$ by a sequence of transformations ... 29

5 Functions 31

Question 5.1 Modulus linear/linear function, range and inverse ... 31
Question 5.2 Self-inverse function, f^{2025} and range of a composite ... 33
Question 5.3 Periodic piecewise function: sketch and area ... 36
Question 5.4 Piecewise function: inverse value, composite existence and f^n ... 38
Question 5.5 Composite with a piecewise function ... 40

1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

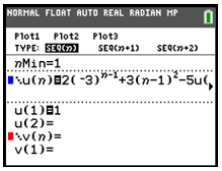
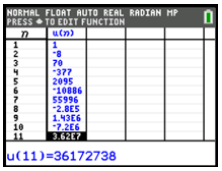
1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu$ $---(1)$ $2+\lambda = 5-2\mu$ $---(2)$ $2\lambda = -8+3\mu$ $---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

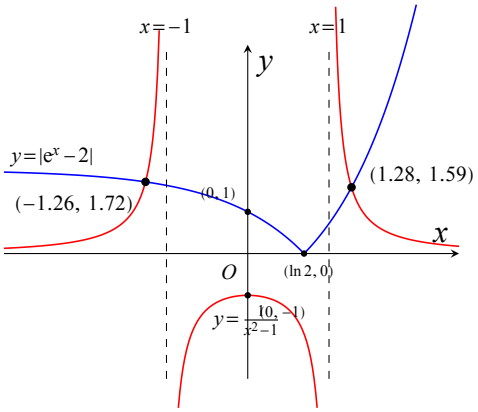
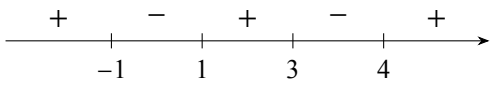
Solution		**Comments**
[4]	Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively. $\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$ $z \geqslant 35$ $\because x \geqslant 0$ $\frac{200}{3}-\frac{5}{3}z \geqslant 0$ $z \leqslant 40$ $\therefore 35 \leqslant z \leqslant 40$ x, y, z are non-negative integers. Using GC table: $z=37 \Rightarrow x=5,\ y=58$ $z=40 \Rightarrow x=0,\ y=60$ $\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$	

2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

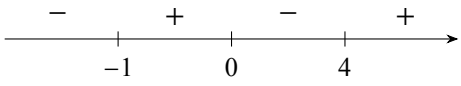
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$  $\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$	

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

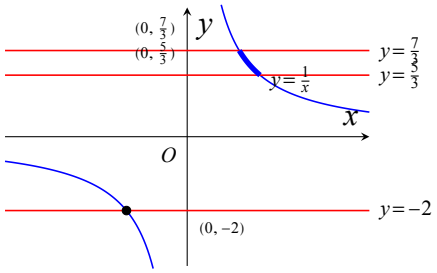
[ACJC 2026 Prelim P1 Q1]

Solution		**Comments**
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1) \cap (2)$: $\therefore x<-1$ or $0<x \leqslant 3$	

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

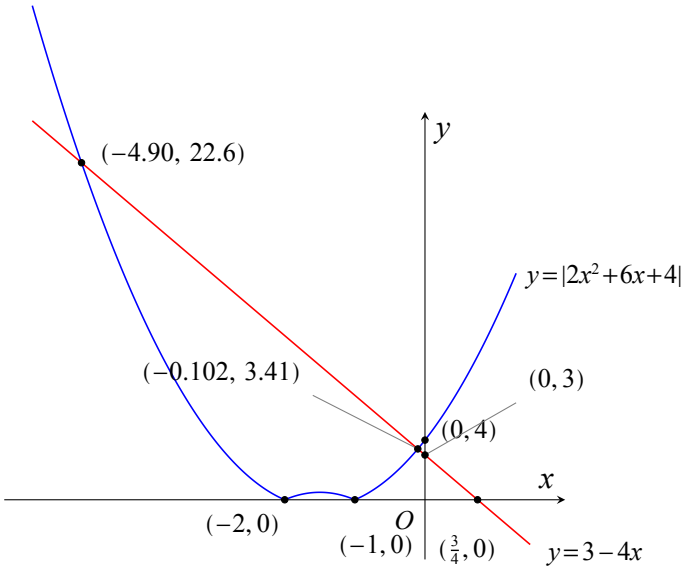
[SAJC 2025 Prelim P2 Q1]

Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2$: $0 \leqslant 0$ (true) $x \neq -2$: $\|x+2\| > 0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

Solution		**Comments**
(i) **[4]**		
	$3-4x=2x^2+6x+4$ $2x^2+10x+1=0$ $x=\dfrac{-5\pm\sqrt{23}}{2}$ $\therefore x<\dfrac{-5-\sqrt{23}}{2}$ or $x>\dfrac{-5+\sqrt{23}}{2}$	
(ii) **[3]**	$\text{Area}=\int_{-3}^{-1}(3-4x-\lvert2x^2+6x+4\rvert)\,dx$ $=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$ $=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$ $=\left[-\frac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\frac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$ $=\left(\frac{16}{3}-20+2\right)-(18-45+3)+\left(-\frac{2}{3}+1-7\right)-\left(-\frac{16}{3}+4-14\right)$ $=20$	

3 Graphing Techniques

3.1 A curve C has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

(a) Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

(b) Sketch C, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

[VJC 2025 Promo Q9(a), (b)]

(a) **[3]**

$$y=\frac{x^2-4x+1}{x+1}$$

$$yx+y=x^2-4x+1$$

$$x^2+(-y-4)x+(1-y)=0$$

No real solution for x: $(-y-4)^2-4(1-y)<0$

$$y^2+8y+16-4+4y<0$$

$$y^2+12y+12<0$$

$$(y+6)^2-24<0$$

$$\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$$

$$-6-2\sqrt{6}<y<-6+2\sqrt{6}$$

$\therefore$ y cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$

(b) **[4]**

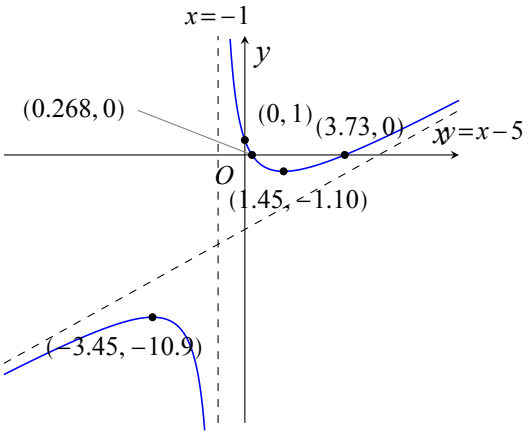
$$y=x-5+\frac{6}{x+1}$$

Asymptotes: $x=-1,\ y=x-5$

$x=0$: $y=1$

$y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$

Turning points: $(1.45, -1.10),\ (-3.45, -10.9)$



3.2 A curve C has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where a and b are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

(a) If C has no stationary points, use differentiation to find the relationship between a and b. [3]

(b) It is now given that $b=a$.

(i) Find the x-coordinates of the turning points. [2]

(ii) Hence sketch C, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

(iii) Verify that the point of intersection of the two asymptotes of C lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of k in terms of a, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

[SAJC 2025 Promo Q7]

Solution		Comments
(a) **[3]**	$\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$ $\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$ $(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$ No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$ Since $a>0,\quad 2b-a<0$ $\therefore a>2b$	
(b)(i) **[2]**	$b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$ $x+3=1$ or -1 $x=-2$ or -4	

(b)(ii)
[3]

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

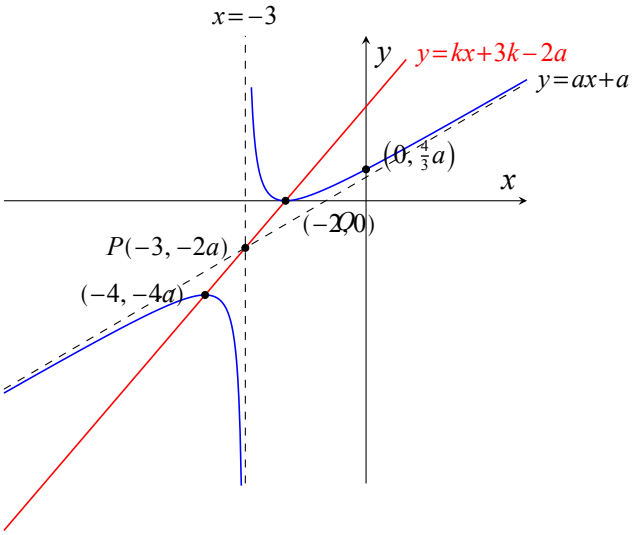
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$



(b)(iii)
[2]

$x = -3$ and $y = ax + a : \quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3 : \quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through P meets $C \Rightarrow k > a$

3.3 A curve C has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and k is a non-zero real constant.

It is given that the function f is strictly increasing.

(a) Find the range of values of k. [3]

(b) Sketch the curve C, giving the coordinates of any points where C crosses the x- and y-axes and the equations of any asymptotes. [3]

(c) By adding a suitable curve to the graph of the curve C in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

[DHS 2025 Promo Q8]

Solution		**Comments**
(a) **[3]**	$\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$ $\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$ f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$ $3x^2+12x+12-k>0 \quad \because (x+2)^2>0$ for $x>-2$ $3(x+2)^2-k>0$ Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$ $k\neq 0 \Rightarrow k<0$	
(b) **[3]**	$y=3x+\dfrac{k}{x+2}$ Asymptotes: $y=3x,\ x=-2$ $x=0:\ y=\dfrac{k}{2}$ $y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$ $x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$	

(c)
[3]

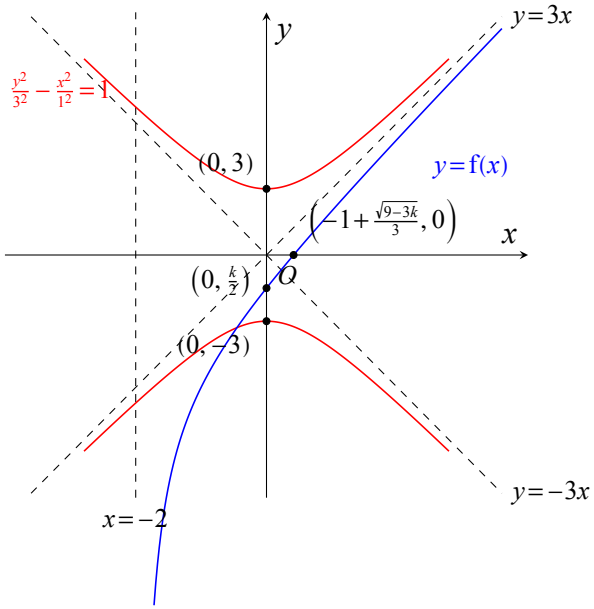
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$



$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

3.4 The curve C is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

(a) Sketch the curve C, labelling the exact coordinates of any axial intercept(s). [1]

(b) Find the equation of the normal to C at the point P where $t=1$. [2]

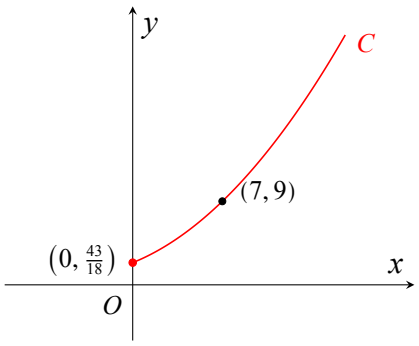
(c) Find the acute angle between the normal to C at the point P where $t=1$ and the tangent to C at the point M where $t=3$. [2]

(d) The curve C is translated 5 units in the positive x-direction and translated 3 units in the negative y-direction, to form the curve D. Find the equation of D in parametric form. [2]

The curve E is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

(e) Show that at the point of intersection of the curves C and E, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

[RVHS 2025 Prelim P1 Q12]

Solution		**Comments**
(a) **[1]**	$t=\frac{5}{6}$: $x=6\left(\frac{5}{6}\right)-5=0$, $y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$ Endpoint (y-intercept): $\left(0, \frac{43}{18}\right)$  	

(b) **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

(c) **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

(d) **[2]**

$$x-5=6t-5 \Rightarrow x=6t$$

$$y+3=2t^2+1 \Rightarrow y=2t^2-2$$

D: $\quad x=6t,\ y=2t^2-2$

(e) **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

3.5 The parametric equations of the curve C are

$$x = \tan\theta - 1 \quad \text{and} \quad y = 2\sec\theta, \quad \text{where } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ, leaving your answer in a single trigonometric function. [2]

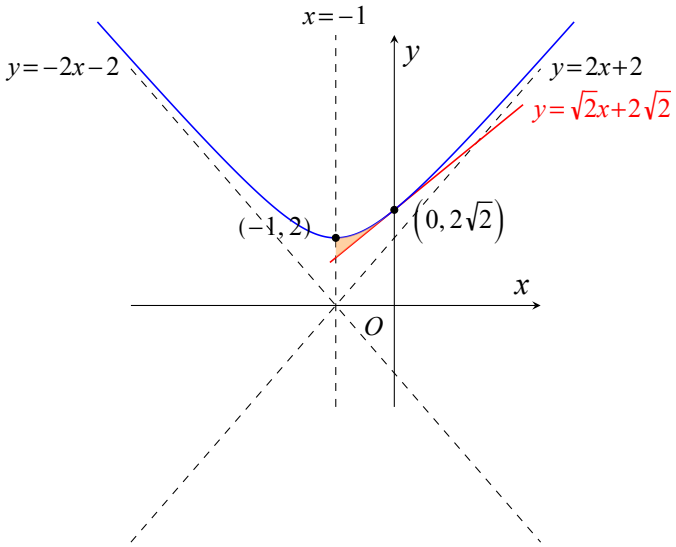
(b) Find the equation of the tangent to C at the point where it cuts the y-axis, leaving your answer in exact form. [3]

(c) Find the Cartesian equation of C. [1]

(d) Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

(e) The region bounded by the tangent found in part (b), the curve C and the line $x = -1$ is rotated completely about the x-axis. Calculate the volume obtained. [3]

[HCI 2024 Promo Q12]

Solution		Comments
(a) **[2]**	$\dfrac{dy}{dx} = \dfrac{dy}{d\theta} \div \dfrac{dx}{d\theta} = \dfrac{2\sec\theta\tan\theta}{\sec^2\theta} = 2\sin\theta$	
(b) **[3]**	$x = 0$: $\tan\theta - 1 = 0 \Rightarrow \tan\theta = 1$ $\theta = \dfrac{\pi}{4}$ since $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$ $\theta = \dfrac{\pi}{4}$: $y = \dfrac{2}{\cos\frac{\pi}{4}} = 2\sqrt{2}$, $\dfrac{dy}{dx} = 2\sin\dfrac{\pi}{4} = \sqrt{2}$ Tangent: $y = \sqrt{2}x + 2\sqrt{2}$	
(c) **[1]**	$\tan^2\theta + 1 = \sec^2\theta$ $(x+1)^2 + 1 = \left(\dfrac{y}{2}\right)^2$ $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2} \Rightarrow 0 < \cos\theta \leqslant 1 \Rightarrow \sec\theta \geqslant 1$ $\therefore (x+1)^2 + 1 = \left(\dfrac{y}{2}\right)^2, \quad y \geqslant 2$	
(d) **[3]**	Asymptotes: $\left(\dfrac{y}{2}\right)^2 = (x+1)^2$ $\dfrac{y}{2} = \pm(x+1)$ $y = 2x+2$ or $y = -2x-2$ Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$  	
(e) **[3]**	$\dfrac{y^2}{4} = (x+1)^2 + 1$ $y^2 = 4(x+1)^2 + 4$ Volume $= \pi\displaystyle\int_{-1}^{0}\left[4\left((x+1)^2+1\right) - \left(\sqrt{2}x + 2\sqrt{2}\right)^2\right]dx$ $= 2.09$ units^3	

Common Question Types

- Prove algebraically that y cannot lie between two values (range of values y cannot take) 3.1
- Sketch a rational curve, stating asymptotes, turning points and axial intercepts 3.1, 3.2, 3.3
- Condition for no stationary points, using differentiation 3.2
- Find the x-coordinates of the turning points 3.2
- Line through the intersection of the asymptotes: range of values of k for real roots 3.2
- Range of values of a constant for a strictly increasing function 3.3
- Add a suitable curve (conic) to the graph to deduce the number of real roots 3.3
- Sketch a parametric curve with an endpoint and axial intercepts 3.4
- Tangent or normal to a parametric curve; acute angle between two lines 3.4, 3.5
- Translate a curve and give its parametric equations 3.4
- Intersection of two parametric curves: show an equation, deduce one point of intersection 3.4
- $\frac{\mathrm{d}y}{\mathrm{d}x}$ of a parametric curve in terms of the parameter 3.5
- Cartesian equation of a parametric curve, then its oblique asymptotes and turning point 3.5
- Volume of revolution of a region bounded by the curve and a tangent 3.5

4 Graphing Transformations

4.1 A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

Step 1: A translation of 2 units in the positive direction of the x-axis.

Step 2: A scaling parallel to the x-axis by factor of 3.

Step 3: A reflection in the y-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

[NJC 2024 Promo Q4(b)]

(b) **[3]**

$$y=-\frac{x^2}{6x+45}$$

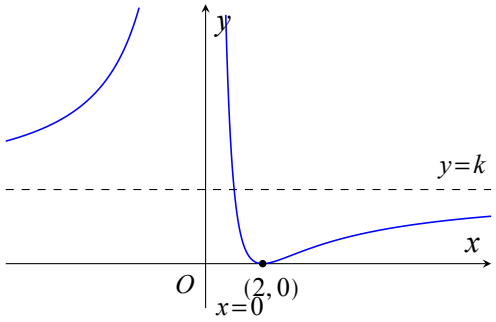
$$(x\to -x):\quad y=-\frac{(-x)^2}{6(-x)+45}=-\frac{x^2}{-6x+45}$$

$$(x\to 3x):\quad y=-\frac{(3x)^2}{-6(3x)+45}=-\frac{9x^2}{-18x+45}=-\frac{x^2}{-2x+5}$$

$$(x\to x+2):\quad y=-\frac{(x+2)^2}{-2(x+2)+5}=-\frac{(x+2)^2}{-2x+1}$$

$$\mathrm{h}(x)=\frac{(x+2)^2}{2x-1}$$

4.2 The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



(a) State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

(b) $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

(c) $y=-\mathrm{f}(|x|)$. [3]

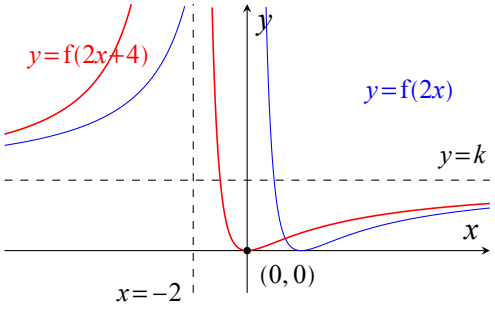
[EJC 2025 Prelim P1 Q4]

(a) **[3]**

$$f(2x+4) = f(2(x+2)) \qquad (x \to x+2)$$

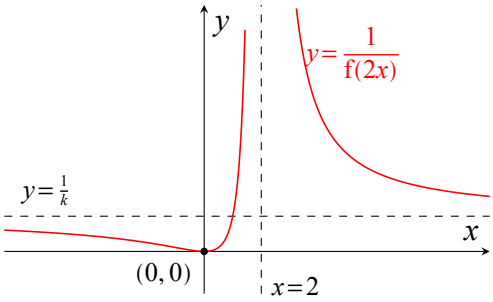
Translation of 2 units in the negative x-direction

$$(2, 0) \to (0, 0), \qquad x = 0 \to x = -2$$



(b) **[3]**

$$(2, 0) \to x = 2, \qquad x = 0 \to (0, 0), \qquad y = k \to y = \frac{1}{k}$$

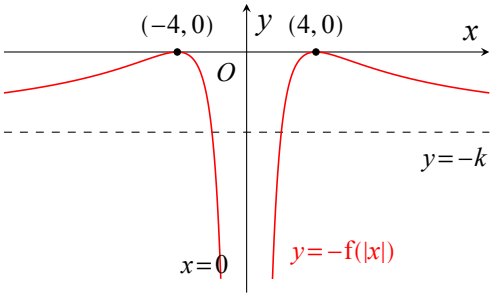


(c) **[3]**

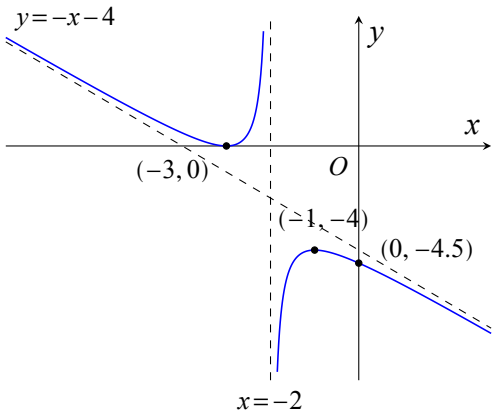
$$f(2x) \xrightarrow{x \to \frac{x}{2}} f(x) \xrightarrow{x \to |x|} f(|x|) \xrightarrow{y \to -y} -f(|x|)$$

$$(2, 0) \to (4, 0) \to (\pm 4, 0) \to (\pm 4, 0)$$

$$y = k \to y = -k, \qquad x = 0$$



4.3



The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the x-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the y-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

(a) $y = \mathrm{f}'(x)$, [3]

(b) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(c) $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

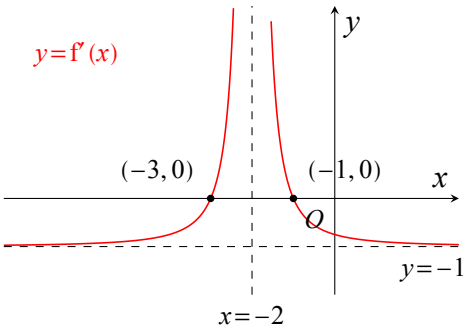
[CJC 2025 Promo Q8]

(a) **[3]**

Turning points $x=-3,\ -1 \Rightarrow (-3,0),\ (-1,0)$

$y=-x-4 \Rightarrow y=-1$

$x=-2 \Rightarrow x=-2$



(b) **[3]**

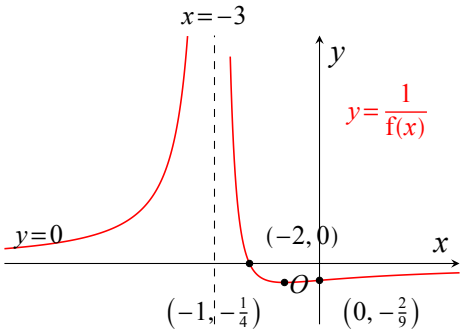
$(-3,0) \Rightarrow x=-3$

$x=-2 \Rightarrow (-2,0)$

$(-1,-4) \Rightarrow \left(-1,-\frac{1}{4}\right)$

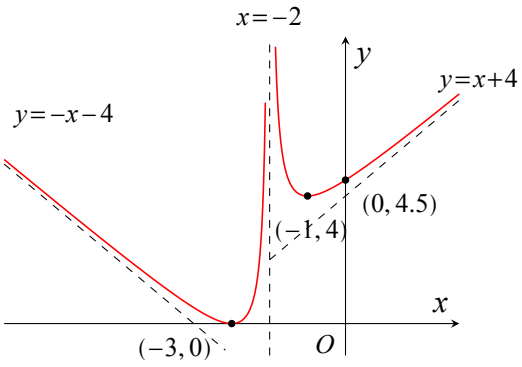
$(0,-4.5) \Rightarrow \left(0,-\frac{2}{9}\right)$

$y=-x-4 \Rightarrow y=0$



(c) **[3]**

$$(-1, -4) \Rightarrow (-1, 4)$$
$$(0, -4.5) \Rightarrow (0, 4.5)$$
$$y = -x - 4 \ (x > -2) \Rightarrow y = x + 4$$



4.4 Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where a and b are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

[RVHS 2026 Prelim P1 Q10(b)]

(b) **[5]**

$$\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \qquad (a=3,\ b=1)$$

$$y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$$

$$\xrightarrow{y\to -y} y=\frac{1}{3x+2}$$

$$\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$$

$$\xrightarrow{x\to x-2} y=\frac{1}{x}$$

1. Translate 3 units in the negative y-direction
2. Reflect about the x-axis
3. Scale by factor 3 parallel to the x-axis
4. Translate 2 units in the positive x-direction

5 Functions

5.1 The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

(b) Hence state the range of f. [1]

(c) Explain why the function f^2 does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where a is a constant.

(d) State the greatest value of a such that the function f^{-1} exists. [1]

(e) Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

[2023 A-Level P1 Q7]

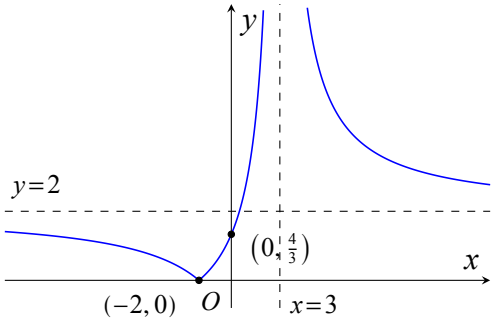
(a) **[3]**

$$y=\left|\frac{2x+4}{3-x}\right|=\left|-2+\frac{10}{3-x}\right|$$

Asymptotes: $y=|-2|=2, \quad x=3$

$$x=0: \quad y=\left|\frac{4}{3}\right|=\frac{4}{3}$$

$$y=0: \quad x=-2$$



(b) **[1]**

$$\mathrm{R_f}=[0,\infty)$$

(c) **[1]**

$$\mathrm{R_f}=[0,\infty), \quad \mathrm{D_f}=(-\infty,3)\cup(3,\infty)$$

$$3\in\mathrm{R_f},\ 3\notin\mathrm{D_f} \Rightarrow \mathrm{R_f}\not\subseteq\mathrm{D_f}$$

$\therefore$ f^2 does not exist

(d) **[1]**

$$a=-2$$

(e) **[4]**

$$x\leqslant -2: \quad \mathrm{f}(x)=\left|\frac{2x+4}{3-x}\right|=-\frac{2x+4}{3-x}$$

$$y=\mathrm{f}(x)$$

$$-3y+xy=2x+4$$

$$x(y-2)=4+3y$$

$$x=\frac{4+3y}{y-2}$$

$$\mathrm{f}^{-1}(x)=\frac{4+3x}{x-2}, \qquad \mathrm{D_{f^{-1}}}=[0,2)$$

5.2 The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

(a) Find the range of f in terms of a. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

(b) Find the value of a. [2]

(c) Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

(d) Find the range of fg. [2]

[HCI 2025 Prelim P1 Q8]

(a) **[1]**

Horizontal asymptote: $y=\frac{a}{5}$

$$\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$$

(b) **[2]**

$$y=\frac{ax-3}{5x-4}$$
$$5xy-4y=ax-3$$
$$5xy-ax=4y-3$$
$$x=\frac{4y-3}{5y-a}$$
$$\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$$
$$\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$$

(c) **[2]**

$$\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$$
$$\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$$
$$=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$$
$$=\cdots$$
$$=\mathrm{f}(2)$$
$$=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$$

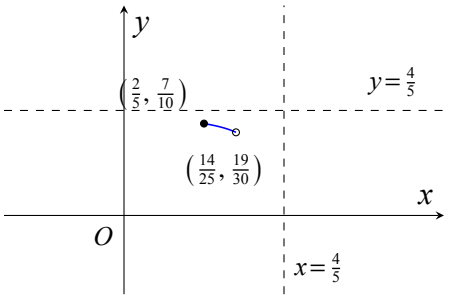
(d) **[2]**

$$R_g = \left[\frac{2}{5}, \frac{14}{25}\right)$$

$$f(x) = \frac{4x-3}{5x-4}$$

$$\left(0, \frac{3}{5}\right) \xrightarrow{g} \left[\frac{2}{5}, \frac{14}{25}\right) \xrightarrow{f} \left(\frac{19}{30}, \frac{7}{10}\right]$$

$$R_{fg} = \left(\frac{19}{30}, \frac{7}{10}\right]$$



5.3 The function f is given by

$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

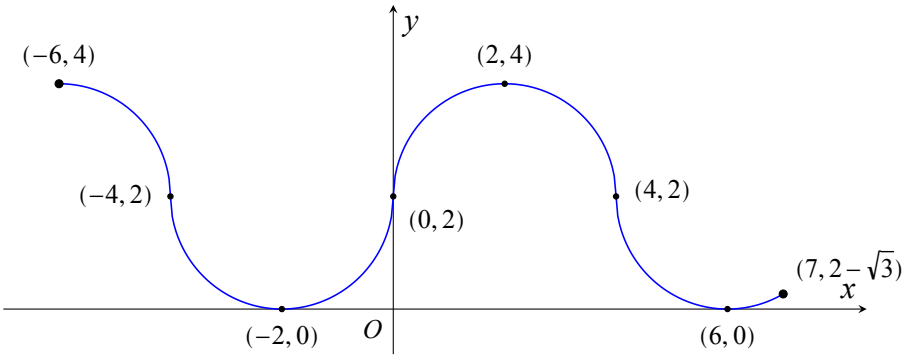
It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

[VJC 2025 Prelim P1 Q5(a), (b)]

(a) **[3]**



(b) **[1]**

$$\text{Area} = 2\times 4 + \frac{1}{2}\pi(2^2) = 8 + 2\pi$$

5.4 The function f is defined by

$$\mathrm{f}(x)=\begin{cases}2-\ln(x+\mathrm{e}) & \text{for } -\mathrm{e}<x<0,\\ \cos\left(\dfrac{\pi}{2}x\right) & \text{for } 0\leqslant x\leqslant 1.\end{cases}$$

(a) Given that the inverse function f^{-1} exists, find the exact value of m if $\mathrm{f}^{-1}(m)=-1$. [2]

Another function g is defined by

$$\mathrm{g}(x)=\sin x, \qquad \text{for } 0\leqslant x\leqslant \frac{\pi}{4}.$$

(b) Explain why the composite function fg exists. [2]

(c) Find the exact value of $\mathrm{fg}\left(\dfrac{\pi}{6}\right)$. [2]

The domain of f is further restricted to $0\leqslant x\leqslant 1$.

(d) By considering $\mathrm{f}^2(1)$ and $\mathrm{f}^3(1)$, deduce the value(s) of $\mathrm{f}^n(1)$ where n is a positive integer. [2]

[HCI 2024 Promo Q8]

(a) **[2]**

$$\begin{aligned} f^{-1}(m) &= -1 \\ m &= f(-1) \\ &= 2 - \ln(e-1) \end{aligned}$$

(b) **[2]**

$$R_g = \left[0, \frac{1}{\sqrt{2}}\right]$$

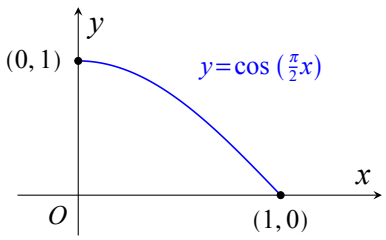
$$D_f = (-e, 1]$$

$$R_g \subseteq D_f \Rightarrow \text{ fg exists}$$

(c) **[2]**

$$\begin{aligned} fg\left(\frac{\pi}{6}\right) &= f\left(\frac{1}{2}\right) \\ &= \cos\left(\frac{\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

(d) **[2]**



$$f^2(1) = f[f(1)] = f(0) = 1$$

$$f^3(1) = f[f^2(1)] = f(1) = 0$$

$$f^n(1) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even} \end{cases}$$

5.5 The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

(i) Find the set of possible values of a such that f^{-1} exist. [1]

(ii) If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

(iii) The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

[ACJC 2022 Prelim P2 Q1]

(i) **[1]**

$$a \in \mathbb{R} \setminus \{0\}$$

(ii) **[3]**

$$y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$$

Reflection of the graph of $y = \mathrm{g}(x)$ in the x-axis, followed by scaling by scale factor 2 parallel to the y-axis, followed by translation 1 unit in the positive y-direction.

(iii) **[3]**

$$\mathrm{gg}(x) = \begin{cases} 7 - (x^2 + 3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7 - (7 - x) & \text{for} \quad 2 < x < 5 \\ (7 - x)^2 + 3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$$

$$\therefore \mathrm{gg}(x) = \begin{cases} 4 - x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2 - 14x + 52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$$